

DBSE & DBD WEEK 1 PRACTICAL

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WEEK 1: DDL COMMANDS AND CONSTRAINTS

Aim:

To practice Data Definition Language (DDL) commands and constraints using MySQL Workbench.

Software Used:

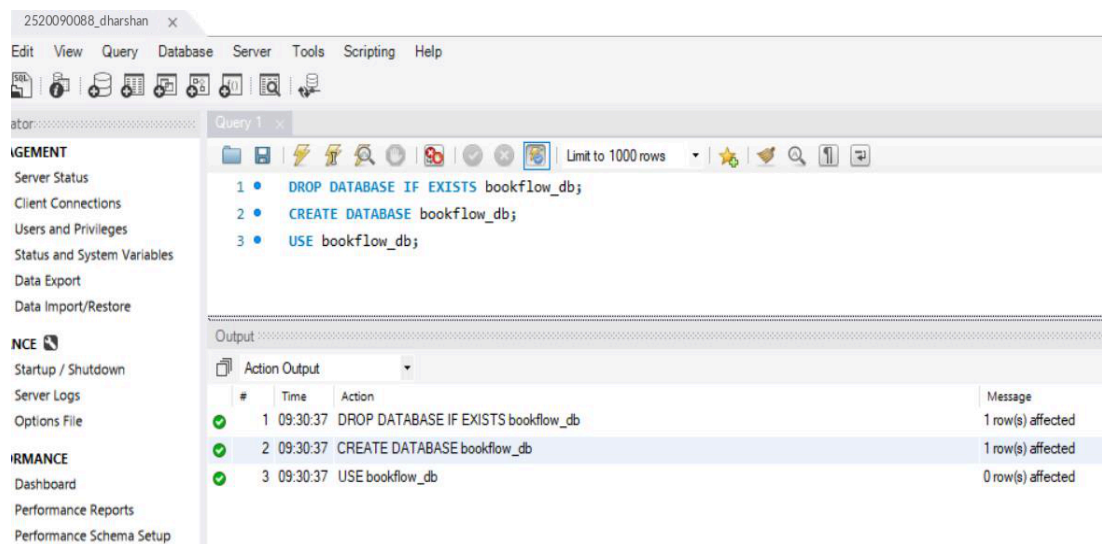
MySQL Workbench

Database Name:

bookflow_db

SQL Operations Performed:

1. Created a database named bookflow_db.



2. Table Creation (DDL) with Constraints

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with the 'Database' folder expanded. The right pane shows the 'Query 1' window with the following SQL script:

```
1 DROP DATABASE IF EXISTS bookflow_db;
2 CREATE DATABASE bookflow_db;
3 USE bookflow_db;
4 CREATE TABLE books (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6     title VARCHAR(255) NOT NULL,
7     isbn VARCHAR(13) NOT NULL UNIQUE,
8     published_year INT,
9     CONSTRAINT chk_published_year CHECK (published_year < 2027)
10 );
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action
1	09:30:37	DROP DATABASE IF EXISTS bookflow_db
2	09:30:37	CREATE DATABASE bookflow_db
3	09:30:37	USE bookflow_db
4	09:40:05	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, i...

3. Create the members table

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with the 'Database' folder expanded. The right pane shows the 'Query 1' window with the following SQL script:

```
3 USE bookflow_db;
4 CREATE TABLE books (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6     title VARCHAR(255) NOT NULL,
7     isbn VARCHAR(13) NOT NULL UNIQUE,
8     published_year INT,
9     CONSTRAINT chk_published_year CHECK (published_year < 2027)
10 );
11 CREATE TABLE members (
12     member_id INT AUTO_INCREMENT PRIMARY KEY,
13     full_name VARCHAR(100) NOT NULL,
14     email VARCHAR(150) NOT NULL UNIQUE
15 );
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action
1	09:30:37	DROP DATABASE IF EXISTS bookflow_db
2	09:30:37	CREATE DATABASE bookflow_db
3	09:30:37	USE bookflow_db
4	09:40:05	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, i...
5	09:44:02	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) N...

4. Inspect the table structures

The screenshot shows the MySQL Workbench interface. The left sidebar contains a tree view with categories: GEMENT, NCE, and RMANCE. The main editor window displays a SQL script for Query 1:

```
10 );
11 CREATE TABLE members (
12     member_id INT AUTO_INCREMENT PRIMARY KEY,
13     full_name VARCHAR(100) NOT NULL,
14     email VARCHAR(150) NOT NULL UNIQUE
15 );
16
17 DESCRIBE books;
18
19 DESCRIBE members;
```

Below the script, the 'Result Grid' shows the structure of the 'members' table:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

The 'Output' pane at the bottom shows the 'Action Output' for the executed queries:

#	Time	Action
1	13:04:29	DROP DATABASE IF EXISTS bookflow_db
2	13:04:29	CREATE DATABASE bookflow_db
3	13:04:29	USE bookflow_db
4	13:04:29	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, ...
5	13:04:29	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) ...
6	13:04:29	DESCRIBE books
7	13:04:29	DESCRIBE members

This screenshot is identical to the one above, showing the same MySQL Workbench interface with the same SQL script and results. The 'Result Grid' and 'Action Output' panes are the same as in the first screenshot.

5. Insert 3 books

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Database' tab with a tree view showing the 'books' table selected. The main query editor shows the following SQL script:

```

15 );
16
17 • DESCRIBE books;
18 • DESCRIBE members;
19
20 • INSERT INTO books (title, isbn, published_year) VALUES
21 ('Cambridge', '9780061122415', 1988),
22 ('Clean Code', '9780132350884', 2008),
23 ('Final war', '9780735211292', 2018);
24 • SELECT * FROM books;

```

The 'Result Grid' shows the output of the 'SELECT * FROM books;' query:

book_id	title	isbn	published_year
1	Cambridge	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Final war	9780735211292	2018
NULL	NULL	NULL	NULL

The 'Output' tab shows the 'Action Output' for the 'books' table, listing the following actions:

- 2 13:10:18 CREATE DATABASE bookflow_db
- 3 13:10:18 USE bookflow_db
- 4 13:10:18 CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL)
- 5 13:10:18 CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL)
- 6 13:10:18 DESCRIBE books
- 7 13:10:18 DESCRIBE members
- 8 13:10:18 INSERT INTO books (title, isbn, published_year) VALUES ('Cambridge', '9780061122415', 1988), ('Clean Code', '9780132350884', 2008), ('Final war', '9780735211292', 2018);
- 9 13:10:18 SELECT * FROM books LIMIT 0, 1000

6. Insert 3 members

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Database' tab with a tree view showing the 'members' table selected. The main query editor shows the following SQL script:

```

21 ('Cambridge', '9780061122415', 1988),
22 ('Clean Code', '9780132350884', 2008),
23 ('Final war', '9780735211292', 2018);
24 • SELECT * FROM books;
25
26 • INSERT INTO members (full_name, email) VALUES
27 ('Sathwika', 'Sathwika@gmail.com'),
28 ('Nikki', 'Nikki@gmail.com'),
29 ('Sanjay', 'Sanjay@gmail.com');
30 • SELECT * FROM members;

```

The 'Result Grid' shows the output of the 'SELECT * FROM members;' query:

member_id	full_name	email
1	Sathwika	Sathwika@gmail.com
2	Nikki	Nikki@gmail.com
3	Sanjay	Sanjay@gmail.com
NULL	NULL	NULL

The 'Output' tab shows the 'Action Output' for the 'members' table, listing the following actions:

- 4 13:15:36 CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL)
- 5 13:15:36 CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL)
- 6 13:15:36 DESCRIBE books
- 7 13:15:36 DESCRIBE members
- 8 13:15:36 INSERT INTO books (title, isbn, published_year) VALUES ('Cambridge', '9780061122415', 1988), ('Clean Code', '9780132350884', 2008), ('Final war', '9780735211292', 2018);
- 9 13:15:36 SELECT * FROM books LIMIT 0, 1000
- 10 13:15:36 INSERT INTO members (full_name, email) VALUES ('Sathwika', 'Sathwika@gmail.com'), ('Nikki', 'Nikki@gmail.com'), ('Sanjay', 'Sanjay@gmail.com');
- 11 13:15:36 SELECT * FROM members LIMIT 0, 1000

7. Tested UNIQUE, NOT NULL and CHECK constraints

The screenshot displays the SQL Workbench interface with a query editor and an output window. The query editor contains the following SQL statements:

```
26 • INSERT INTO members (full_name, email) VALUES
27 ('Sathwika', 'Sathwika@gmail.com'),
28 ('Nikki', 'Nikki@gmail.com'),
29 ('Sanjay', 'Sanjay@gmail.com');
30 • SELECT * FROM members;
31
32 • INSERT INTO books (title, isbn, published_year)
33 VALUES (NULL, '9999999999999', 2010);
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES ('Time Traveler', '8888888888888', 2030);
36 • INSERT INTO members (full_name, email)
37 VALUES ('Srihitha', 'Nikki@gmail.com');
```

The output window shows the results of these queries:

#	Time	Action	Message
4	13:15:36	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL...	0 row(s) affected
5	13:15:36	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100)...	0 row(s) affected
6	13:15:36	DESCRIBE books	4 row(s) returned
7	13:15:36	DESCRIBE members	3 row(s) returned
8	13:15:36	INSERT INTO books (title, isbn, published_year) VALUES ('Cambridge', '9780061122415', 1988), ('Clean Cod...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	13:15:36	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
10	13:15:36	INSERT INTO members (full_name, email) VALUES ('Sathwika', 'Sathwika@gmail.com'), ('Nikki', 'Nikki@gmail...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
11	13:15:36	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
12	13:19:51	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key
13	13:26:46	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null